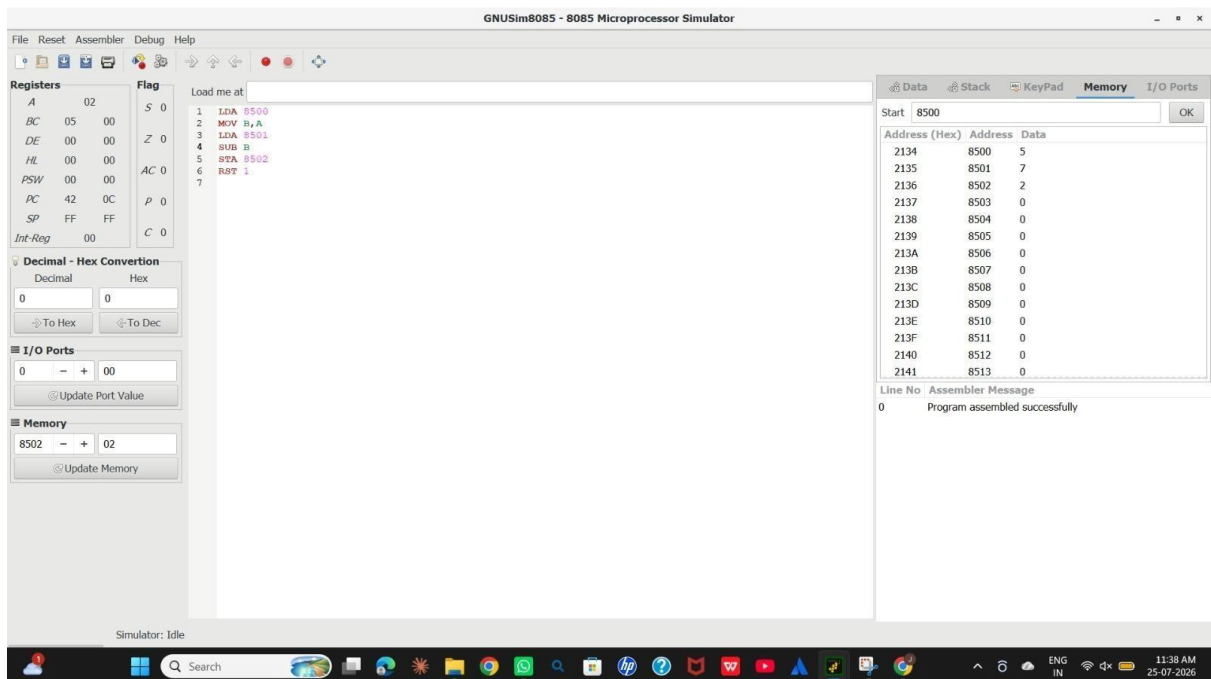
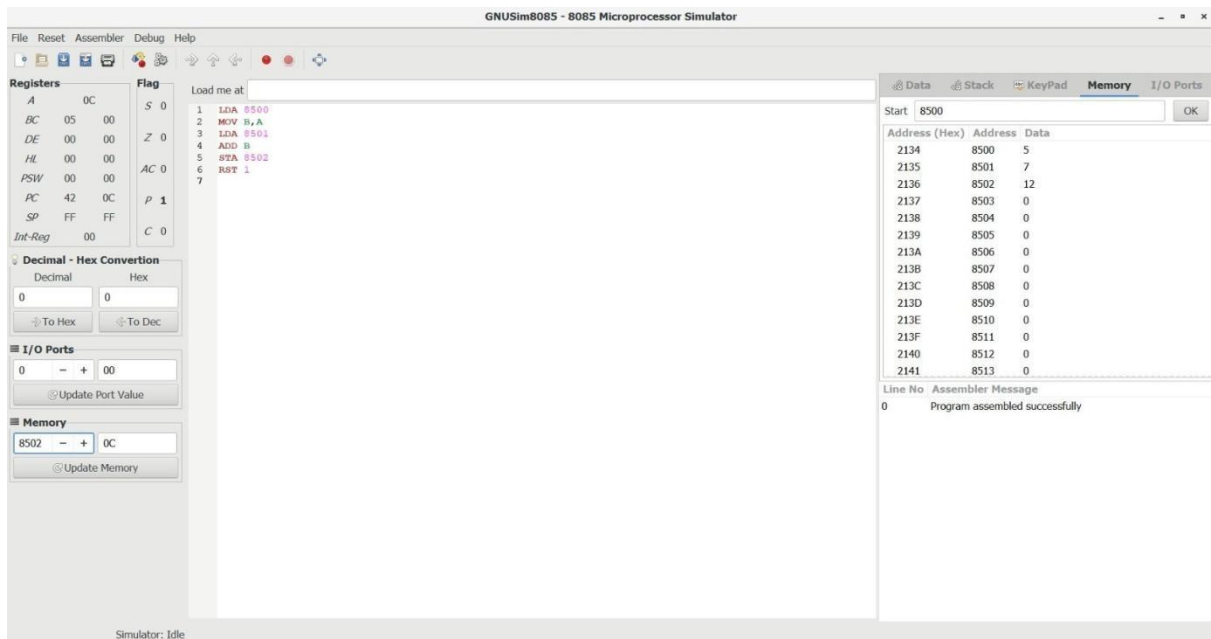
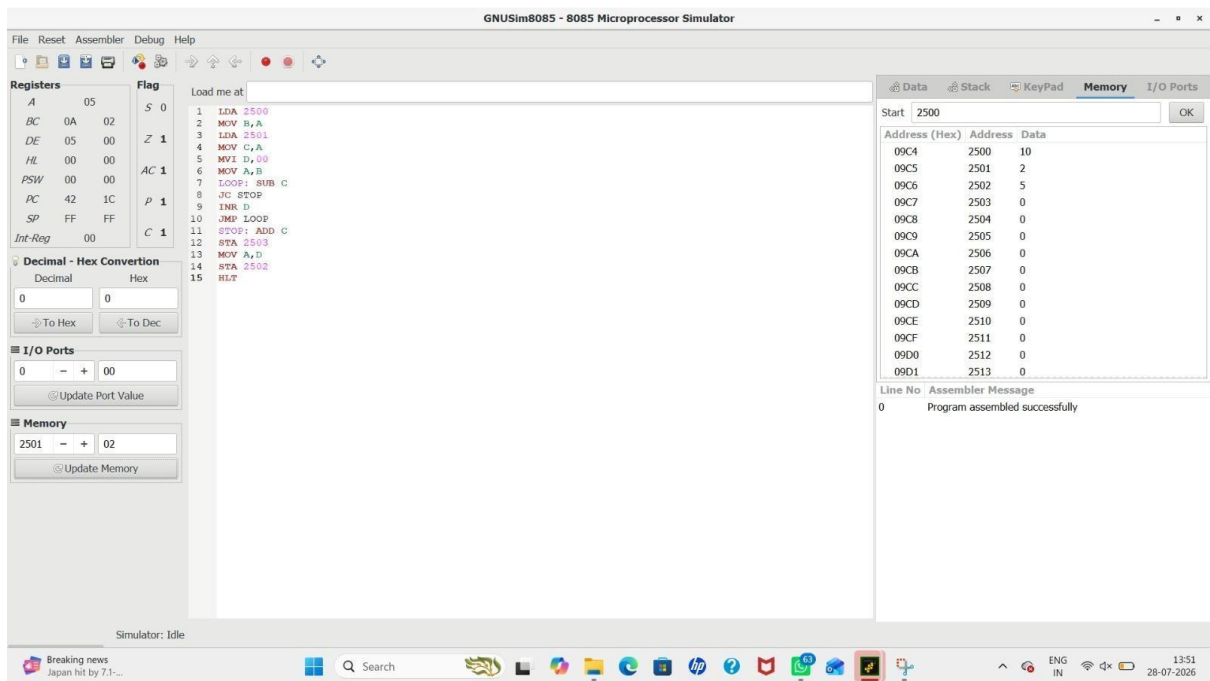
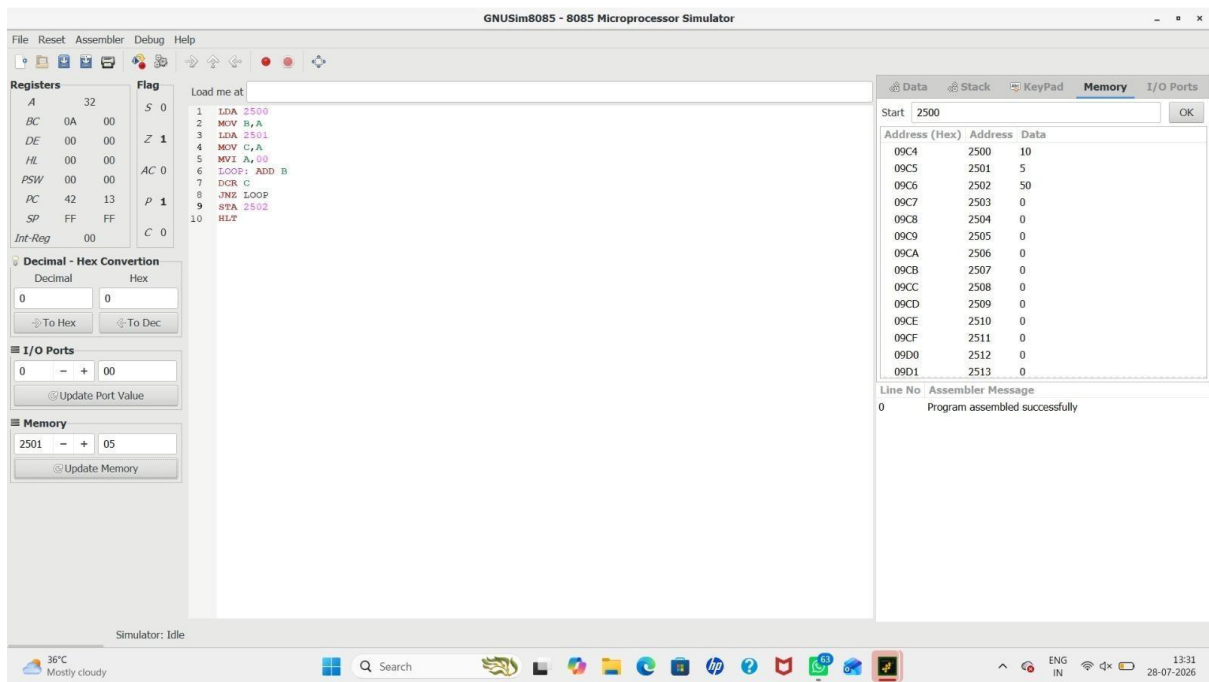






GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	02
BC	05 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

I/O Ports

0 - + 00

Update Port Value

Memory

8502 - + 02

Update Memory

Load me at

```
1 LDA 8500
2 MOV B,A
3 LDA 8501
4 SUB B
5 STA 8502
6 RST 1
7
```

Start 8500

Address (Hex)	Address	Data
2134	8500	5
2135	8501	7
2136	8502	2
2137	8503	0
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0
2140	8512	0
2141	8513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0F
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

I/O Ports

0 - + 00

Update Port Value

Memory

2501 - + 14

Update Memory

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

Start 2500

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	20
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0F
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2501 - + 14

Update Memory

Simulator: Idle

37°C Mostly cloudy

Search

ENG IN

14:15 28-07-2026

Data Stack Keypad Memory I/O Ports

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	20
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	14
BC	14 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	1
Z	0
AC	0
P	0
C	1

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 14

Update Memory

Simulator: Idle

Data Stack Keypad Memory I/O Ports

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	20
09C5	2501	15
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	05	S 1
BC	1E 00	Z 0
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 1
PC	42 14	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2501 - + 05

Update Memory

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC SECOND_SMALLER
6 MOV A,B
7 STA 2502
8 HLT
9 SECOND_SMALLER: STA 2502
10 HLT
```

Start: 2500

Address (Hex)	Address	Data
09C4	2500	30
09C5	2501	5
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag
A	28	S 1
BC	32 28	Z 0
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 1
PC	42 11	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 28

Update Memory

Load me at

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MOV A,B
6 STA 2501
7 MOV A,C
8 STA 2500
9 HLT
```

Start: 2500

Address (Hex)	Address	Data
09C4	2500	40
09C5	2501	50
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag	Value
A	E3	S	1
BC	32 28	Z	0
DE	00 00	AC	0
HL	00 00	P	1
PSW	00 00	C	1
PC	42 08		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2501 - + E3

Update Memory

Load me at

```

1 LDA 2500
2 CMA
3 STA 2501
4 HLT

```

Start 2500

Address (Hex)	Address	Data
09C4	2500	28
09C5	2501	227
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No: 0 Assembler Message: Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value	Flag	Value
A	00	S	0
BC	32 28	Z	1
DE	00 00	AC	1
HL	00 00	P	1
PSW	00 00	C	0
PC	42 14		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 18

Update Memory

Load me at

```

1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT

```

Start 2500

Address (Hex)	Address	Data
09C4	2500	24
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No: 0 Assembler Message: Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	01
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 0E
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	1
P	0
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 23

Update Memory

Load me at

```
1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT
```

Data Stack Keypad Memory I/O Ports

Start: 2500

Address (Hex)	Address	Data
09C4	2500	35
09C5	2501	1
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	00
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 19

Update Memory

Load me at

```
1 LDA 2500
2 ORA A
3 JP POSITIVE
4 MVI A, 01H
5 STA 2501
6 HLT
7 POSITIVE: MVI A, 00H
8 STA 2501
9 HLT
```

Data Stack Keypad Memory I/O Ports

Start: 2500

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 01, BC 03 00, DE 00 00, HL 10 6A, PSW 00 00, PC 42 20, SP FF FF, Int-Reg 00. Flag: S 0, Z 1, AC 0, P 1, C 1.

Decimal - Hex Conversion: Decimal 0, Hex 0. To Hex, To Dec.

I/O Ports: 0 - + 00. Update Port Value.

Memory: 4205 - + 09. Update Memory.

Load me at:

```

1 LXI H, 4200
2 MOV C, M
3 DCR C
4 START: MOV D, C
5 LXI H, 4201
6 LOOP: MOV A, M
7 INX H
8 CMP M
9 JC SKIP
10 JZ SKIP
11 MOV B, M
12 MOV M, A
13 DCX H
14 MOV M, B
15 INX H
16 SKIP: DCR D
17 JNZ LOOP
18 DCR C
19 JNZ START
20 HLT

```

Start 4200 OK

Address (Hex)	Address	Data
1068	4200	5
1069	4201	1
106A	4202	3
106B	4203	6
106C	4204	7
106D	4205	9
106E	4206	1
106F	4207	4
1070	4208	6
1071	4209	7
1072	4210	9
1073	4211	0
1074	4212	0
1075	4213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 09, BC 09 00, DE 00 00, HL 10 6A, PSW 00 00, PC 42 20, SP FF FF, Int-Reg 00. Flag: S 0, Z 1, AC 0, P 1, C 0.

Decimal - Hex Conversion: Decimal 0, Hex 0. To Hex, To Dec.

I/O Ports: 0 - + 00. Update Port Value.

Memory: 4205 - + 01. Update Memory.

Load me at:

```

1 LXI H, 4200
2 MOV C, M
3 DCR C
4 START: MOV D, C
5 LXI H, 4201
6 LOOP: MOV A, M
7 INX H
8 CMP M
9 JNC SKIP
10 JZ SKIP
11 MOV B, M
12 MOV M, A
13 DCX H
14 MOV M, B
15 INX H
16 SKIP: DCR D
17 JNZ LOOP
18 DCR C
19 JNZ START
20 HLT

```

Start 4200 OK

Address (Hex)	Address	Data
1068	4200	5
1069	4201	9
106A	4202	7
106B	4203	6
106C	4204	3
106D	4205	1
106E	4206	1
106F	4207	4
1070	4208	6
1071	4209	7
1072	4210	9
1073	4211	0
1074	4212	0
1075	4213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	1C
BC	09 00
DE	00 00
HL	10 6D
PSW	00 00
PC	42 18
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Load me at

```

1 LXI H,4200
2 MOV C,M
3 DCR C
4 INX H
5
6 MOV A,M
7
8 BACK: INX H
9 CMP M
10 JC UPDATE
11 JMP NEXT
12
13 UPDATE: MOV A,M
14 NEXT: DCR C
15 JNZ BACK
16
17 STA 4300
18 HLT

```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

4205 - + 06

Update Memory

Simulator: Idle

Data Stack KeyPad Memory I/O Ports

Start 4300 OK

Address (Hex)	Address	Data
10CC	4300	28
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

Line No Assembler Message

0 Program assembled successfully

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	03
BC	09 00
DE	00 00
HL	10 6D
PSW	00 00
PC	42 18
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	1

Load me at

```

1 LXI H,4200
2 MOV C,M
3 DCR C
4 INX H
5
6 MOV A,M
7
8 BACK: INX H
9 CMP M
10 JC SKIP
11 JZ SKIP
12 MOV A,M
13
14 SKIP: DCR C
15 JNZ BACK
16 STA 4300
17 HLT

```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

4209 - + 00

Update Memory

Simulator: Idle

Data Stack KeyPad Memory I/O Ports

Start 4300 OK

Address (Hex)	Address	Data
10CC	4300	3
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

Line No Assembler Message

0 Program assembled successfully

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	02
BC	03 00
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

4201 - + 02

Update Memory

Load me at

```
1 LDA 4200
2 STA 4300
3
4 LDA 4201
5 MOV E,A
6
7 CHECK: LDA 4300
8 MOV D,A
9 DIVIDE: MOV A,D
10 CMP E
11 JC NOT_DIVISIBLE
12 JZ FOUND
13 SUB E
14 MOV D,A
15 JMP DIVIDE
16 NOT_DIVISIBLE: LDA 4300
17 MOV B,A
18 LDA 4200
19 ADD B
20 STA 4300
21 JMP CHECK
22 FOUND: HLT
```

Start 4300 OK

Address (Hex)	Address	Data
10CC	4300	6
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	01
BC	01 01
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

4200 - + 03

Update Memory

Load me at

```
1 LDA 4200
2 MOV B,A
3 LDA 4201
4 MOV C,A
5 LOOP: MOV A,B
6 ORA A
7 JZ BZERO
8 MOV A,C
9 ORA A
10 JZ CZERO
11 MOV A,B
12 CMP C
13 JC CGREATER
14 JZ EQUAL
15 SUB C
16 MOV B,A
17 JMP LOOP
18 CGREATER: MOV A,C
19 SUB B
20 MOV C,A
21 JMP LOOP
22 EQUAL: MOV A,B
23 STA 4205
24 HLT
25 BZERO: MOV A,C
26 STA 4205
27 HLT
28 CZERO: MOV A,B
29 STA 4205
30 HLT
```

Start 4200 OK

Address (Hex)	Address	Data
1068	4200	3
1069	4201	2
106A	4202	0
106B	4203	0
106C	4204	0
106D	4205	1
106E	4206	0
106F	4207	0
1070	4208	0
1071	4209	0
1072	4210	0
1073	4211	0
1074	4212	0
1075	4213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	78
BC	78 00
DE	06 78
HL	00 00
PSW	00 00
PC	42 28
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

Decimal - Hex Conversion

Decimal: 0 Hex: 0

I/O Ports

Memory

4200 - + 05

Update Memory

Load me at

```

1 LDA 4200
2 MOV C,A
3
4 MVI B,01
5 MVI D,02
6
7 CHECK: LDA 4200
8 CMP D
9 JC DONE
10 JZ MULTIPLY
11
12 MULTIPLY: MOV A,D
13 MOV C,A
14 MVI E,00
15
16 ADD_LOOP: MOV A,E
17 ADD B
18 MOV B,A
19 DCR C
20 JNZ ADD_LOOP
21
22 MOV A,E
23 MOV B,A
24
25 INR D
26 JMP CHECK
27
28 DONE: MOV A,B
29 STA 4300
30 HLT

```

Memory

Address (Hex)	Address	Data
10CC	4300	120
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	A9
BC	90 09
DE	A0 00
HL	00 00
PSW	00 00
PC	42 20
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	1
Z	0
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

I/O Ports

Memory

4200 - + 19

Update Memory

Load me at

```

1 LDA 4200
2 MOV B,A
3
4 ANI 0FFH
5 MOV C,A
6
7 MOV A,B
8 ANI 0FFH
9
10 RRC
11 RRC
12 RRC
13 RRC
14
15 MOV B,A
16
17 MVI D,00
18 MVI E,0AH
19
20 TEN_TIMES: MOV A,D
21 ADD B
22 MOV D,A
23 DCR E
24 JNZ TEN_TIMES
25 MOV A,D
26 ADD C
27 STA 4205
28
29 HLT

```

Memory

Address (Hex)	Address	Data
1068	4200	25
1069	4201	2
106A	4202	0
106B	4203	0
106C	4204	0
106D	4205	169
106E	4206	0
106F	4207	0
1070	4208	0
1071	4209	0
1072	4210	0
1073	4211	0
1074	4212	0
1075	4213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle